

Rushill Nair

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SUMMARY

AI/ML Engineer with hands-on experience building end-to-end machine learning systems, including time-series forecasting, feature engineering, model training, tuning, and evaluation. Worked on RAG-based Generative AI systems with vector databases and LLM integration, familiar with backend development using FastAPI and SQL for data-driven applications.

EXPERIENCE (KPIs)

Data Science GET – AI Division | Mahindra Group

Aug 2025 – Feb 2026

- Trained ensemble models on 10M+ sensor points to predict tyre forces across terrains, enabling engineers to reduce physical testing cycles by 3 months through a **SMAPE reduction from 14% to 3%**.
- Engineered robust features using statistical methods, FFT analysis, and vehicle dynamics modeling, enabling reliable performance across diverse terrains for the ML model and **reducing instrumentation and trial costs by INR 2.5 lakh per day**.
- Built FastAPI-based Python APIs and monitoring dashboards to track post-production model health and performance for internal stakeholders.

Applied AI Intern | LiquidMindAI

Feb 2025 – May 2025

- Built a RAG pipelines that **increased retrieval precision from 78% to 89%** for trade compliance document analysis.
- Experimented various retrieval and re-ranking methods to handle specialized legal terminology, delivering rapid Streamlit prototypes to accelerate stakeholder feedback loops.

Machine Learning Research Intern | Indian Institute of Science (ICWaR)

Apr 2024 – Feb 2025

- Trained multi-horizon forecasting models (TFT, SARIMAX) that halved daily error rates for flood discharge compared to static modelling.
- Used Monte Carlo simulations for probabilistic uncertainty quantification, providing 95% confidence intervals for risk-aware planning.

PROJECTS (TECHNICAL IMPLEMENTATION)

ForceCast (Virtual Sensor System) | Mahindra Group

Python, XGBoost, Optuna, FastAPI

- Worked on ForceCast, an ensemble ML method to estimate wheel forces and vehicle dynamics during early-stage trials (T1–T2), reducing reliance on WFT instrumentation in low-precision scenarios.
- Engineered signal processing and feature pipelines for denoising, temporal alignment, and extraction of dynamics-aware feature(CAN, accelerometers, suspension signals) from multi-rate sensor streams.
- Introduced iterative feedback-based refinement loops to improve prediction consistency across varying terrains and driving conditions, trained models on GPU-enabled GCP Vertex AI and optimized hyperparameters using Optuna to maximize generalization across models.

AgriRAG (IEEE Publication) | NMIT Research

BERT, LSTMs, PyTorch, LangChain

- Built an hybrid AI system for climate-resilient agricultural forecasting, combining statistical models (SARIMAX, Prophet, NARX) and deep learning architectures (LSTM, BiLSTM with attention) to predict yields for major crops across climate-vulnerable districts in Karnataka.
- Integrated a domain-specific Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and the o3-mini model(via Azure) to ground predictions in government reports and research literature, enabling context-aware, explainable decision support.
- Implemented self-reflection loops where the LLM critiques and verifies its outputs against retrieved evidence, improving factual consistency and reducing hallucinations in advisory recommendations.

SKILLS

- ML/DL:** Scikit-Learn, XGBoost, Numpy, Pandas, Optuna, SHAP, Tensorflow-Keras, Plotly
- GenAI:** RAG designs, LangChain, Vector DBs (FAISS, Qdrant), Re-ranking, RAGAS, HuggingFace
- Programming/Backend:** Python, SQL, FastAPI, Docker, MLflow, GCP Vertex AI, Streamlit.

EDUCATION

Nitte Meenakshi Institute of Technology (NMIT)

2021 – 2025

B.E. in Electronics and Communication Engineering

Bangalore, India

ACHIEVEMENTS & EXTRACURRICULARS

- Music:** Semi-professional drummer; ranked Top 10 at Mood Indigo (IITB) and Waves (BITS Goa).